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Real-Time Muscle Biomechanical Modeling with Physics-Informed Neural Networks for Strain-Constrained Path Planning in Robot-assisted Pelvic Fracture Reduction

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Abstract— **In robot-assisted pelvic fracture reduction, path planning constitutes a fundamental component, and real-time prediction of the biomechanical response of soft tissues, including deformation and maximum stress, is critical for the accuracy and safety of the procedure. To address this challenge, Muscle-PINNs is proposed as a physics-informed neural network (PINN) for skeletal muscle biomechanics that integrates physical priors with data. First, a five-parameter, second-order polynomial hyperelastic constitutive law is adopted, with the balance of linear momentum and boundary conditions embedded as mesh-free PDE-residual constraints; FEA data are used for supervision to generate displacement fields that are physically consistent and observation compliant. Second, these displacement fields are incorporated into RRT* with a strain-threshold constraint and a length–strain node cost: high-strain branches are pruned during node expansion, followed by path optimization. Experimental results demonstrate that Muscle-PINNs achieves millisecond-level inference ($9 \sim 10$ ms), providing a substantial speed advantage over conventional FEA while maintaining accuracy in predicting displacements and maximum principal strains under diverse loading conditions. This approach enables real-time, physics-driven decision support for minimally invasive pelvic fracture reduction, balancing computational efficiency and mechanical fidelity.**

I. Introduction

The pelvic ring is composed of the hip (innominate) bones, sacrum, and coccyx, and serves to transmit body weight, protect the viscera, and provide sites for muscular attachment. Pelvic fractures are most often caused by high-energy trauma and account for 3%–8% [1]. Unstable Tile B/C patterns are frequently associated with massive hemorrhage and visceral injury, with a mortality of about 30% [2]. With the development of science and technology, robot-assisted minimally invasive reduction has been applied to pelvic fractures, offering the advantages of reduced tissue trauma, lower infection rates, and faster recovery.

Owing to the extensive peripelvic musculature, the reduction forces applied by the robot deform the muscles. The resulting reactive forces can compromise reduction accuracy, and excessive manipulation may cause muscle injury or tearing. Accordingly, it is necessary to systematically assess, from a biomechanical perspective, the effects of pelvic reduction on surrounding soft tissues such as muscles. Giraudet *et al*. reconstructed a three-dimensional pelvic muscle model from MRI, providing a foundation for teaching and for understanding normal and pathological anatomy [3]. Ead *et al*. reconstructed the injured hemipelvis from relevant CT data, offering a reference for reduction-surgery planning [4]. However, existing models predominantly focus on osseous structures and treat soft tissues (e.g., muscles and ligaments) in an overly simplified manner, resulting in deviations from physiological reality.

Soft tissues commonly exhibit anisotropy, heterogeneity, viscoelasticity, hyperelasticity, and strong nonlinearity. Existing biomechanical modeling approaches largely fall into two categories, the first of which is finite-element analysis (FEA) grounded in constitutive laws. For example, Phillips *et al*. treated the peripelvic muscles and ligaments as fixed constraints to compute pelvic stress–strain fields [5]; Lei *et al*. built a fractured-pelvis musculoskeletal finite-element model(FEM) based on a five-parameter Mooney–Rivlin law to simulate muscle deformation and stress–strain responses during reduction [6]. Although FEM can accurately reproduce muscular responses during reduction, its modeling and computational costs are high and each scenario is time-consuming, which challenges the real-time requirements of pelvic reduction path planning. The second approach involves constructing a musculoskeletal dynamic model using OpenSim. However, this method remains time-consuming and costly.

The emergence of physics-informed neural networks (PINNs) has provided a new approach for soft tissue biomechanical modeling. The PINNs embed physical priors, such as governing equations and boundary conditions, into neural networks to solve differential equations in a mesh free manner, and have shown significant promise in fluid, solid, and heat-transfer applications [7]–[9]. In biomechanics, Martín *et al*. coupled PINNs with cardiac electrophysiology to propose EP-PINNs, which accurately reconstruct action potentials and

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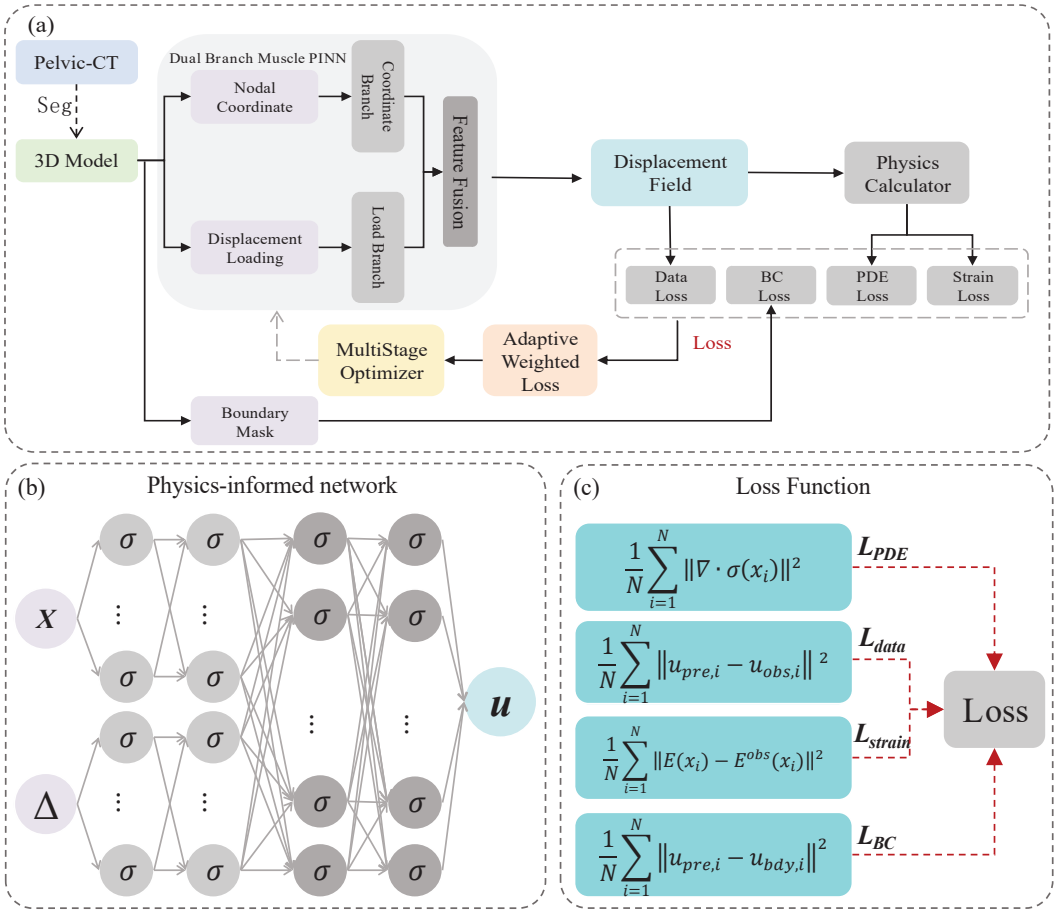


Fig. 1. Muscle-PINNs biomechanics modeling algorithm

infer key parameters [10]. Zhang *et al*. employed PINNs to model blood-flow evolution, achieving a mapping from the spatiotemporal domain to four-dimensional hemodynamics [11].

In soft tissue modeling, electromyography-based data-driven approaches (EMG) have been investigated to learn mappings between kinematic variables and neuromuscular states [12]–[15]. Zhang *et al*. proposed a physics-informed deep learning framework for musculoskeletal modeling that predicts muscle forces and joint motion from surface EMG [16]. Building on this work, they further developed a deep learning framework to estimate muscle forces directly from unlabeled surface EMG signals [17].

To address the challenges of muscle modeling during pelvic fracture reduction, a physics-informed neural network (PINN)-based algorithm for skeletal muscle biomechanics is presented, enabling rapid prediction of peripelvic muscle deformation within the path planning workflow for pelvic reduction. The principal contributions of this study are summarized as follows.

- A physics-informed neural networks (PINNs) based scheme for skeletal-muscle biomechanics is proposed, in which physical priors are fused with data. The architecture employs a space–load–decoupled feature extractor: dedicated branches encode geometric topology and loading conditions, and their representations are fused at the feature level rather than through a single-stream input, thereby improving generalization across diverse operating scenarios.
- The Muscle-PINNs model is integrated with an RRT* path planner. The predicted displacement fields are used to compute muscle strain, enforce strain thresholds during tree expansion, and optimize the final route via a strain-aware cost, thereby effectively avoiding excessive muscular strain while yielding feasible, near-optimal pelvic reduction paths.

II. Method

This study proposes a physics-informed neural network framework for modeling the mechanics of skeletal muscle soft tissue, as illustrated in Fig. 1a. By tightly integrating governing physical laws with deep learning, the framework enables high fidelity simulation of complex biomechanical behavior in soft tissues.

A. Biomechanical Modeling of Muscle Tissue

Skeletal muscle is a strongly nonlinear soft material whose mechanical response is dominated by hyperelasticity; under physiological loading it can undergo large deformations with near recoverability and exhibits stress softening, hysteresis,

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and anisotropy. In biomechanical modeling, the choice of constitutive law is critical. In engineering practice, muscle is often idealized as an isotropic hyperelastic solid and represented by invariant-based strain–energy formulations (e.g., Mooney–Rivlin, Ogden, Yeoh), from which the stress–strain relations are derived via the strain–energy density W[18]. Given the small inter-node displacements and slow, controllable motion in pelvic reduction path planning, a quasi-static analysis is adopted to describe muscle stretching.

Given the biomechanical characteristics of muscle tissue, we adopt a second-order polynomial hyperelastic model, in which the strain–energy density W is additively decomposed into deviatoric and volumetric parts.

$$W = W_{\text{dev}} + W_{\text{vol}} \tag{1}$$

The deviatoric strain energy density W_{dev} is formulated in terms of the first and second invariants, I_1 and I_2, of the right Cauchy–Green deformation tensor $\mathbf{C}$.

$$\begin{aligned} W_{\text{dev}} &= C_{10}(I_1-3) + C_{01}(I_2-3) + C_{20}(I_1-3)^2 \\ &\quad + C_{11}(I_1-3)(I_2-3) + C_{02}(I_2-3)^2 \end{aligned} \tag{2}$$

Here, C_{10} and C_{01} denote the stiffness under small strains, while C_{20}, C_{11}, and C_{02} are parameters that capture nonlinear behavior and complex deformation.

The volumetric term W_{vol} accounts for near-incompressibility, and W_{vol} is expressed as.

$$W_{\text{vol}} = \frac{2}{D_1}(J-1)^2 + \frac{4}{D_2}(J-1)^4 \tag{3}$$

Here, D_1 and D_2 are incompressibility (bulk-penalty) parameters, and $J = \det\mathbf{F}$ denotes the determinant of the deformation gradient $\mathbf{F}$.

The biomechanical modeling of muscle tissue is formulated within a continuum–mechanics framework. Under quasi-static conditions and in the absence of body forces, the system satisfies the balance of linear momentum. The derivation starts from the deformation mapping.

$$\boldsymbol{\varphi} : \mathbf{X} \mapsto \mathbf{x} = \mathbf{X} + \mathbf{u}(\mathbf{X}) \tag{4}$$

with $\mathbf{X}$ the material (reference) coordinates, $\mathbf{x}$ the spatial (current) coordinates, and $\mathbf{u}$ the displacement field. The deformation gradient tensor $\mathbf{F}$ is defined as.

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \mathbf{I} + \nabla_{\mathbf{X}}\mathbf{u} \tag{5}$$

For hyperelastic materials, the strain–energy density W is taken to depend on the isochoric (modified) invariants $\bar{I}_1, \bar{I}_2$ and on the Jacobian J. The second Piola–Kirchhoff stress tensor $\mathbf{S}$ is derived from W as

$$\mathbf{S} = 2\frac{\partial W}{\partial \mathbf{C}} = 2\frac{\partial W}{\partial \bar{I}_1}\frac{\partial \bar{I}_1}{\partial \mathbf{C}} + 2\frac{\partial W}{\partial \bar{I}_2}\frac{\partial \bar{I}_2}{\partial \mathbf{C}} + 2\frac{\partial W}{\partial J}\frac{\partial J}{\partial \mathbf{C}} \tag{6}$$

where $\mathbf{C} = \mathbf{F}^{\mathsf{T}}\mathbf{F}$ is the right Cauchy–Green tensor and $J = \det\mathbf{F}$. Useful derivatives include

$$\begin{cases} I_1 = \text{tr}(\mathbf{C}), \\ I_2 = \dfrac{1}{2}\left[(\text{tr}\,\mathbf{C})^2 - \text{tr}\left(\mathbf{C}^2\right)\right] \end{cases} \tag{7}$$

The second Piola–Kirchhoff stress can be mapped to the Cauchy stress as[19].

$$\boldsymbol{\sigma} = J^{-1}\mathbf{F}\mathbf{S}\mathbf{F}^{\mathsf{T}} \tag{8}$$

and, under quasi-static conditions, the balance of linear momentum in the current (spatial) configuration reads

$$\nabla\cdot\boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \tag{9}$$

where $\mathbf{b}$ is the body-force density (so that $\nabla\cdot\boldsymbol{\sigma} = \mathbf{0}$ when body forces are neglected)[20].

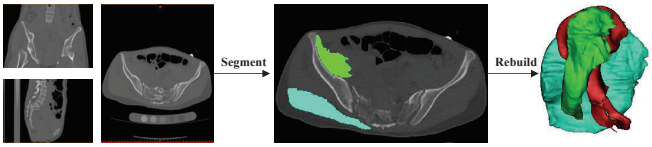


Fig. 2. Three-dimensional muscle modeling

B. Network Architecture and Training of Muscle-PINNs

To accurately predict the displacement response of muscle tissue under multiple loading scenarios, we propose a dual-branch physics-informed neural network (Muscle-PINNs). As illustrated in Fig. 1b, Muscle-PINNs consists of a coordinate branch, a load branch, and a fusion (merge) network. The coordinate branch processes the spatial inputs (X, Y, Z) with two hidden layers; the load branch processes the loading inputs (L_x, L_y, L_z), also with two hidden layers. The outputs of the two branches are concatenated and passed to the fusion network, which produces the normalized displacement vector $\mathbf{u} = [u_x, u_y, u_z]^{\mathsf{T}}$. The forward pass is given by

$$\mathbf{u}_{\text{pred}} = f_{\text{merge}}(\text{concat}(f_{\text{coord}}(\mathbf{X}), f_{\text{load}}(\mathbf{L}))) \tag{10}$$

Here, f denotes a multilayer perceptron (MLP) in which each layer applies an affine transformation followed by a tanh activation. This dual-branch design enables the model to capture geometric and load-dependent effects separately, thereby improving adaptability and generalization across multiple operating conditions. Material parameters are embedded as fixed constants within the network to enforce physical consistency.

The physics module computes the deformation gradient and stresses via automatic differentiation. For each mini-batch, PyTorch autograd evaluates the components of $\partial\mathbf{u}/\partial\mathbf{X}$ componentwise; the resulting gradients are rescaled by the normalization factors to recover physical units. Using the hyperelastic strain–energy function, the modified invariants $\bar{I}_1$ and $\bar{I}_2$ are computed, the isochoric stress $\boldsymbol{\sigma}_{\text{iso}}$ is assembled, and the volumetric contribution $\boldsymbol{\sigma}_{\text{vol}}$ is added. The Green–Lagrange strain tensor $\mathbf{E}$ is then vectorized into a six-dimensional representation for the strain-fitting loss.

The total loss of the Muscle-PINNs model comprises data fitting, boundary condition, equilibrium, and strain terms, as illustrated in Figure 1c. Adaptive weights dynamically balance these components using EMA-smoothed mixing of inverse and proportional weights, bounded weight ranges, warmup, and a stagnation fallback strategy to adjust their

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relative contributions. The overall objective is defined as a five-term weighted sum.

$$L_{\text{total}} = \sum_{k=1}^{5} \lambda_k L_k \tag{11}$$

where the data-fitting loss term is defined as:

$$L_{\text{data}} = \frac{1}{|N|} \sum_{i \in N} \left\| \hat{\mathbf{u}}_i - \hat{\mathbf{u}}_i^{\text{obs}} \right\|^2 \tag{12}$$

Fixed-boundary loss term.

$$L_{\text{fixed}} = \frac{1}{|N|} \sum_{i \in N} \| \hat{\mathbf{u}}_i \|^2 \tag{13}$$

Load-boundary loss term.

$$L_{\text{stretch}} = \frac{1}{|N|} \sum_{i \in N} \left\| \hat{\mathbf{u}}_i - \hat{\mathbf{L}} \right\|^2 \tag{14}$$

Momentum-balance (equilibrium) loss term.

$$L_{\text{PDE}} = \frac{1}{|N|} \sum_{i \in N} \| \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}_i) \|^2 \tag{15}$$

Strain-fitting loss term.

$$L_{\text{strain}} = \frac{1}{|N|} \sum_{i \in N} \left\| \mathbf{E}(\mathbf{x}_i) - \mathbf{E}^{\text{obs}}(\mathbf{x}_i) \right\|^2 \tag{16}$$

Optimization is conducted in two stages. Adam is used initially for rapid convergence. After a preset number of epochs, the optimizer switches to L-BFGS for refined optimization.

C. Pelvic Fracture Reduction Path Planning

The Muscle–PINNs model is incorporated into the classical RRT* path-planning framework to accommodate muscle-strain constraints. Specifically, RRT* incrementally connects the start and goal by randomly sampling the search space and extending the tree, while Muscle–PINNs efficiently predicts the muscle strain at each candidate node to determine node feasibility.

In each iteration, the algorithm first randomly samples a point in the configuration space and identifies the nearest node in the existing tree. The line segment between them is interpolated to form a new candidate node; if this segment collides with environmental obstacles, the node is discarded, otherwise it is temporarily added to the tree. Then, for each new node, a rigid-body transformation maps the node's position to the initial configuration of the adhesion point—denoting the adhesion point's initial coordinates by $\mathbf{C}_{\text{init}}$, the rotation matrix by $\mathbf{R}$, and the translation vector by $\mathbf{T}$. The transformed coordinates are.

$$\mathbf{C}_{\text{new}} = \mathbf{R}\,\mathbf{C}_{\text{init}} + \mathbf{T} \tag{17}$$

The displacement values at the muscle attachment points are supplied to Muscle-PINNs; from the model-predicted displacement field the deformation gradient is computed, and the Cauchy (infinitesimal) strain—the symmetric part of the displacement gradient—is obtained as.

$$\mathbf{E}_i = \tfrac{1}{2}\left(\mathbf{F}_i^{\mathsf{T}}\mathbf{F}_i - \mathbf{I}\right) \tag{18}$$

The maximum principal strain at each point was obtained through eigenvalue decomposition, with the peak value recorded and returned to the RRT* path-planning module.

The muscle-strain constraint checks whether the maximum principal strain at the candidate path node exceeds a prescribed threshold, defined as the maximum principal strain at the goal point; nodes that violate this threshold are discarded.

III. Experiments and Results

A. Data Acquisition and Preprocessing

To ensure the physical fidelity and accuracy of the data, the dataset used in this study was derived from simulation outputs of a pelvic musculoskeletal model generated using FEA software. The specific preprocessing workflow is outlined as follows.

- **3D Model Reconstruction and Assembly:** As shown in Fig. 2, CT scans of a pelvic fracture patient served as the raw data. Gray-scale threshold segmentation was performed in segmentation software to extract muscle and bone tissues, yielding three-dimensional geometric models in STL format. To meet FEA mesh-quality requirements, the muscle STL files were repaired, denoised, and surface-reconstructed. The resulting STP files were then precisely assembled to construct a complete pelvis–muscle coupled model.
- **Finite Element Simulation Setup:** In the FEA software, the muscle domain was assigned a five-parameter second-order polynomial hyperelastic constitutive law. The mesh consisted of hybrid ten-node tetrahedral elements. The basal boundary of the muscle was fixed, and a prescribed displacement was applied to the muscle–ilium attachment region. The displacement field **U** and strain field **E** were then computed.
- **Data Integration and Normalization:** Global node coordinates, fixed-boundary nodes (zero displacement), and load-boundary nodes with prescribed displacements Δ were extracted. During preprocessing, fixed, loading, and interior masks were generated via coordinate matching. Spatial coordinates and displacement loads were concatenated into a six-dimensional input vector. Features were scaled to $[-1, 1]$ using MinMaxScaler, while displacement targets were normalized independently to preserve physical scale and aid convergence. The resulting multi-scenario dataset comprised tensorized features, targets, masks, and associated scalers, improving data consistency and enabling training across diverse loading conditions.

B. Muscle Stretching Simulation

The experimental muscle model comprised the gluteus ectogluteus and the iliacus segmented from patient CT images, consistent with canonical biomechanical characteristics. As shown in Fig. 3b, the basal surface of each muscle was fully fixed, and a prescribed displacement was applied at the muscle–ilium attachment points. The material behavior was represented by a five-parameter, second-order polynomial hyperelastic law.

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According to the literature, Lemmon *et al*. experimentally evaluated the mechanical behavior of a five-parameter Mooney–Rivlin muscle model and determined the material parameter values [21], as shown in Table. I.

TABLE I
MODEL PARAMETER

Parameter	C_{10}	C_{01}	C_{20}	C_{11}	C_{02}	D_1	D_2
Value	0.08556	0.05841	0.039	0.02319	0.00851	4.37e-6	6.811e-7

The dataset was generated via quasi-static tensile FEA under displacement control of the gluteus ectogluteus and iliacus muscles. Muscle-PINNs were trained on 100 simulated cases of the gluteus ectogluteus and iliacus. During inference, with boundary and loading conditions identical to the FEA, the model took muscle node coordinates and displacement increments (L_x, L_y, L_z) as inputs to predict displacement fields for all three muscles. The displacement contour comparison in Fig. 4 shows close agreement between Muscle-PINNs and FEA, accurately capturing muscle deformation under displacement-controlled loading. Quantitative

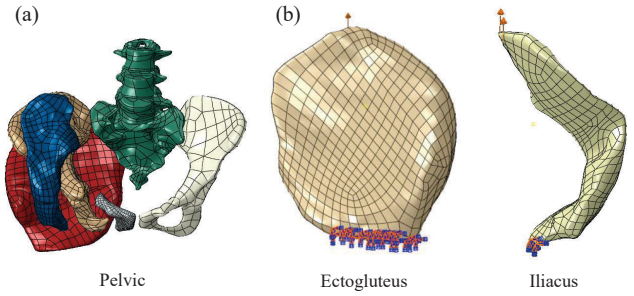


Fig. 3. Boundary conditions for muscles

evaluation use two metrics: the mean absolute error (MAE) of displacement and the relative error of the maximum principal strain $E_{\max}$. Table. II reports results for test case 9, where the prescribed displacement load for the gluteus maximus is $(8.096, -1.578, -0.283)$ and for the iliacus is $(12.74, -4.28, 0.59)$.

TABLE II
CONTRAST EXPERIMENT

Method	Muscle	MAE (mm)	$E_{\max}$ error
Muscle-PINNs	Gluteus maximus	0.35	0.003
	Iliacus	0.437	0.0006
Pure PINN	Gluteus maximus	0.868	0.145
	Iliacus	0.652	0.026
Data only	Gluteus maximus	0.062	0.189
	Iliacus	0.147	0.045

A comparison of three approaches shows that a purely data-driven model attains low displacement-field error but, lacking physical constraints, exhibits large strain-prediction bias; a purely physics-driven model reduces strain error but sacrifices displacement accuracy and underperforms overall; by fusing physics and data, Muscle-PINNs maintains controlled displacement error while markedly reducing strain error, achieving superior overall performance.

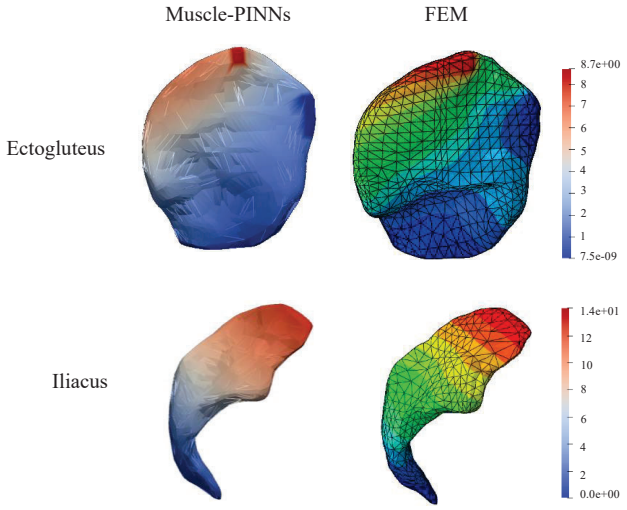


Fig. 4. Comparison of displacement fields: Muscle-PINNs vs. FEM

C. Pelvic Fracture Reduction Path Simulation

To evaluate the effectiveness of Muscle–PINNs in pelvic reduction path planning, a comparison was conducted using conventional finite-element-based muscle simulation as the baseline. Data were obtained from computed tomography (CT) scans of a pelvic fracture patient, from which a three-dimensional pelvic point cloud and the main muscle nodes were reconstructed. A pelvis–muscle coupled finite element model was established in the FEA software, with the gluteus maximus and iliacus on the affected side selected as the subjects of investigation.

Path search employed a standard RRT* with a maximum of 800 iterations. After each node expansion and collision check, the pelvis center-of-mass translation $\mathbf{T}$ and rotation $\mathbf{R}$ were computed; the rigid-body transformation $(\mathbf{R}, \mathbf{T})$ was then applied to all muscle–pelvis attachment points to obtain displacements, which served as inputs to Muscle-PINNs. The predicted displacement field was post-processed to obtain the maximum principal strain $E_{\max}$. Any candidate node whose $E_{\max}$ exceeded that at the goal point was discarded.

Fig. 5 compares the maximum principal strain at each node along the same path: Fig. 5a for the ectogluteus and Fig. 5b for the iliacus,Fig 5c illustrates the RRT* pelvic fracture reduction path based on Muscle-PINNs. Relative to FEM, the mean relative error of $E_{\max}$ with Muscle-PINNs is 0.031 (ectogluteus) and 0.002 (iliacus). In terms of efficiency, FEM requires about $2 \sim 10$ min per displacement case, whereas Muscle-PINNs inference takes only $9 \sim 10$ ms, substantially reducing computational cost.

IV. CONCLUSIONS

This study proposes Muscle-PINNs, a physics-informed neural network for skeletal-muscle biomechanics. By embedding a second-order, five-parameter polynomial hyperelastic

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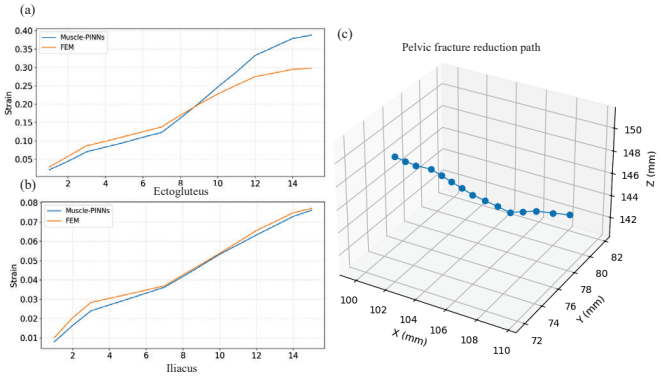


Fig. 5. Comparison of maximum principal stress (pelvic reduction)

constitutive law together with strain-compatibility residuals and other physical constraints into the network, the method enables a mesh-free solution that reduces reliance on large-scale data while preserving physical consistency. The predicted deformation field is integrated into an RRT* path optimizer, where a strain threshold is used to generate safe trajectories that avoid high-strain regions. Experiments show millisecond-level inference, delivering a substantial speedup over conventional FEM and enabling real-time decision support for pelvic fracture reduction, while effectively suppressing muscle strain without compromising path quality. The approach combines efficiency and accuracy, offering a practical, physics-driven solution for pelvic reduction path planning.

Despite the promising results, this study has limitations. First, the accuracy of Muscle-PINNs is lower than that of FEM and depends on high-quality FEM data, which constrains generalizability to clinical settings. Second, the current framework focuses on quasi-static problems and does not account for time-varying factors such as muscle fatigue. Third, the integration with robotic path planning has not yet been validated on physical hardware, introducing sim-to-real transfer risks. Future work will extend the method to dynamic simulations by incorporating time-dependent constraints.

References

[1] M. Mi, N. K. Kanakaris, X. Wu, and P. V. Giannoudis, Management and outcomes of open pelvic fractures: an update, Injury, vol. 52, no. 10, pp. 2738–2745, 2021.

[2] M. Burkhardt, A. Kristen, U. Culemann, et al., Pelvic fracture in multiple trauma: are we still up-to-date with massive fluid resuscitation?, Injury, vol. 45, S70–S75, 2014.

[3] G. Giraudet, L. Patrouix, C. Fontaine, X. Demondion, M. Cosson, and C. Rubod, Three dimensional model of the female perineum and pelvic floor muscles, European Journal of Obstetrics Gynecology and Reproductive Biology, vol. 226, pp. 1–6, 2018.

[4] M. S. Ead, L. Westover, S. Polege, S. McClelland, J. L. Jaremko, and K. K. Duke, Virtual reconstruction of unilateral pelvic fractures by using pelvic symmetry, International Journal of Computer Assisted Radiology and Surgery, vol. 15, no. 8, pp. 1267–1277, 2020.

[5] A. T. M. Phillips, P. Pankaj, C. R. Howie, A. S. Usmani, and A. H. R. W. Simpson, Finite element modelling of the pelvis: inclusion of muscular and ligamentous boundary conditions, Medical Engineering Physics, vol. 29, no. 7, pp. 739–748, 2007.

[6] J. Lei, Y. Li, and H. Xu, Study on circumpelvic muscle deformation and dynamic simulation of pelvic fracture reduction, Computer Methods in Biomechanics and Biomedical Engineering, vol. 26, no. 6, pp. 734–743, 2023.

[7] B. Lütjens, C. H. Crawford, M. Veillette, and D. Newman, PCE-PINNs: physics-informed neural networks for uncertainty propagation in ocean modeling, arXiv preprint arXiv:2105.02939, 2021.

[8] M. Rasht-Behesht, C. Huber, K. Shukla, and G. E. Karniadakis, Physics-informed neural networks (PINNs) for wave propagation and full waveform inversions, Journal of Geophysical Research: Solid Earth, vol. 127, no. 5, pp. e2021JB023120, 2022.

[9] M. Penwarden, S. Zhe, A. Narayan, and R. M. Kirby, Multifidelity modeling for physics-informed neural networks (PINNs), Journal of Computational Physics, vol. 451, pp. 110844, 2022.

[10] C. Herrero Martin, A. Oved, R. A. Chowdhury, E. Ullmann, N. S. Peters, A. A. Bharath, and M. Varela, EP-PINNs: cardiac electrophysiology characterisation using physics-informed neural networks, Frontiers in Cardiovascular Medicine, vol. 8, pp. 768419, 2022.

[11] X. Zhang, B. Mao, Y. Che, J. Kang, M. Luo, A. Qiao, Y. Liu, H. Anzai, M. Ohta, Y. Guo, et al., Physics-informed neural networks (PINNs) for 4D hemodynamics prediction: an investigation of optimal framework based on vascular morphology, Computers in Biology and Medicine, vol. 164, pp. 107287, 2023.

[12] H. Su, W. Qi, Z. Li, Z. Chen, G. Ferrigno, and E. De Momi, Deep neural network approach in EMG-based force estimation for human–robot interaction, IEEE Transactions on Artificial Intelligence, vol. 2, no. 5, pp. 404–412, 2021.

[13] J. Zhang, Z. Ruan, Q. Li, and Z.-Q. Zhang, Toward robust and efficient musculoskeletal modeling using distributed physics-informed deep learning, IEEE Transactions on Instrumentation and Measurement, vol. 72, pp. 1–11, 2023.

[14] C. Wu, Q. Cao, F. Fei, D. Yang, B. Xu, G. Zhang, H. Zeng, and A. Song, Optimal strategy of sEMG feature and measurement position for grasp force estimation, PLOS ONE, vol. 16, no. 3, pp. e0247883, 2021.

[15] T. Bao, S. A. R. Zaidi, S. Xie, P. Yang, and Z.-Q. Zhang, A CNN–LSTM hybrid model for wrist kinematics estimation using surface electromyography, IEEE Transactions on Instrumentation and Measurement, vol. 70, pp. 1–9, 2020.

[16] J. Zhang, Y. Zhao, F. Shone, Z. Li, A. F. Frangi, S. Q. Xie, and Z.-Q. Zhang, Physics-informed deep learning for musculoskeletal modeling: predicting muscle forces and joint kinematics from surface EMG, IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 31, pp. 484–493, 2022.

[17] S. Ma, J. Zhang, C. Shi, P. Di, I. D. Robertson, and Z.-Q. Zhang, Physics-informed deep learning for muscle force prediction with unlabeled sEMG signals, IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 32, pp. 1246–1256, 2024.

[18] S. K. Melly, L. Liu, Y. Liu, and J. Leng, A review on material models for isotropic hyperelasticity, International Journal of Mechanical System Dynamics, vol. 1, no. 1, pp. 71–88, 2021.

[19] G. A. Holzapfel and R. W. Ogden, Modeling the biomechanical properties of soft biological tissues: constitutive theories, European Journal of Mechanics–A/Solids, vol. 112, pp. 105634, 2025.

[20] F. Caforio, F. Regazzoni, S. Pagani, E. Karabelas, C. Augustin, G. Haase, G. Plank, and A. Quarteroni, Physics-informed neural network estimation of material properties in soft tissue nonlinear biomechanical models, Computational Mechanics, vol. 75, no. 2, pp. 487–513, 2025.

[21] D. Lemmon, T.-Y. Shiang, A. Hashmi, J. S. Ulbrecht, and P. R. Cavanagh, The effect of insoles in therapeutic footwear—a finite element approach, Journal of Biomechanics, vol. 30, no. 6, pp. 615–620, 1997.

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